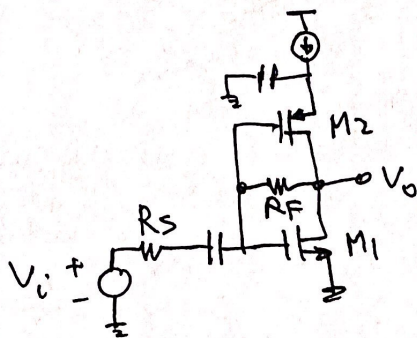


problem 5.7:

- r_o is not neglected
- The LNA is matched at the input.

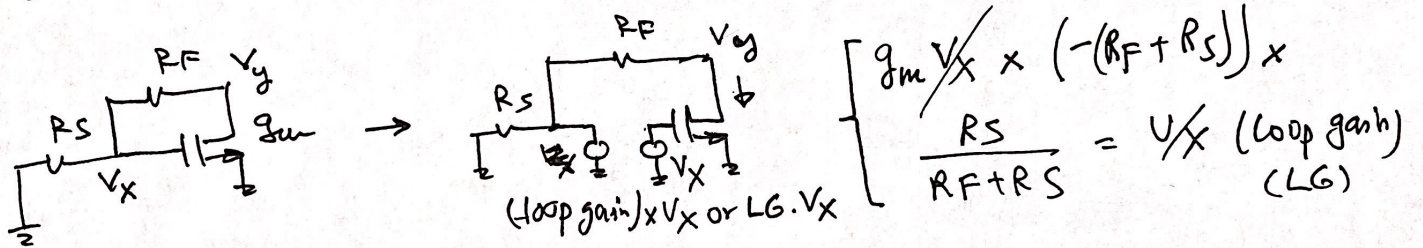


$$g_m = g_{mp} + g_{mn}, \quad r_o = r_{on} \parallel r_{op}$$

loop gain:

→ for the case where there is no r_o effect ($r_o \rightarrow \infty$), loop gain can be

calculated as:

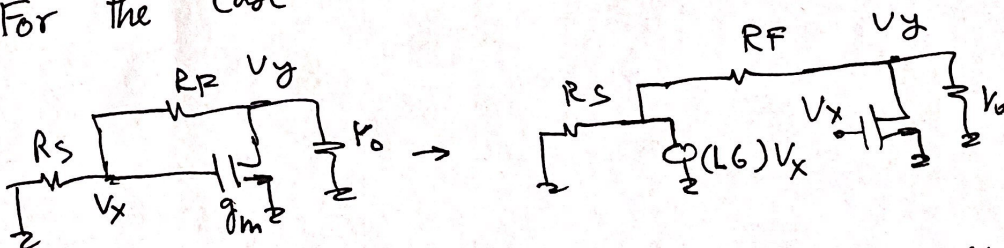


$$\rightarrow -g_m (R_F + R_S) \cdot \frac{R_S}{R_F + R_S} = \text{Loop gain (LG)}$$

$$\rightarrow \text{loop gain} = -g_m R_S, \quad Z_{in} = \frac{1}{g_m} = R_S \text{ (for matching)}$$

$$\rightarrow \boxed{\text{Loop gain}_{(LG)} = -1}$$

→ For the case where $r_o \neq \infty$, loop is given by:

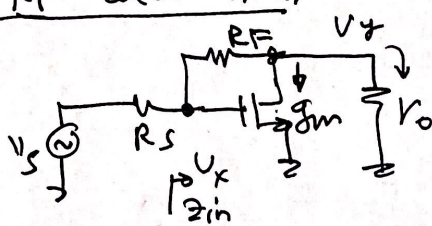


$$-g_m (r_o \parallel (R_F + R_S)) \cdot \frac{R_S}{R_F + R_S} \quad V_x = (LG) V_x$$

$$\rightarrow LG = -g_m (r_o \parallel (R_F + R_S)) \cdot \frac{R_S}{R_F + R_S} = - \frac{g_m r_o R_S}{r_o + R_F + R_S}$$

$$\rightarrow LG = \frac{-g_m R_S}{R_0 + R_F + R_S} \quad \text{if } R_0 \gg R_F, R_S \rightarrow \boxed{LG \approx -g_m R_S} \quad P.2$$

Gain/NF Calculation:



Kcl

$$V_Y \left(\frac{1}{R_0} + \frac{1}{R_F} \right) = -g_m V_X + \frac{V_X}{R_F}$$

$$\left| \frac{V_Y}{V_X} = \frac{-(g_m R_F - 1)}{1 + R_F/R_0} \right|$$

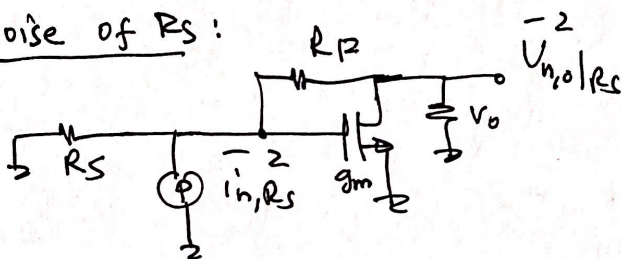
Matching Condition:

$$Z_{in} = \frac{V_X}{(V_X - V_Y)/R_F} = \frac{R_F}{1 - \frac{V_Y}{V_X}} = \frac{R_F}{1 + \frac{g_m R_F - 1}{1 + \frac{R_F}{R_0}}}$$

$$\rightarrow Z_{in} = \frac{(1 + \frac{R_F}{R_0}) R_F}{1 + \frac{R_F}{R_0} + g_m R_F} = \frac{(1 + \frac{R_F}{R_0}) R_F}{R_F (\frac{1}{R_0} + g_m)} \approx \frac{1 + \frac{R_F}{R_0}}{g_m} = R_S$$

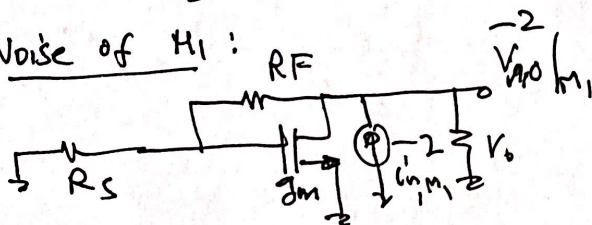
$$\rightarrow \boxed{g_m R_S \approx 1 + \frac{R_F}{R_0}}$$

Noise of R_S :



$$\begin{aligned} \bar{V}_{n,RS}^2 &= \bar{V}_{n,RS}^2 \left(\frac{R_S}{2} \right)^2 \left(\frac{g_m R_F - 1}{1 + \frac{R_F}{R_0}} \right)^2 \\ \text{or } \bar{V}_{n,RS}^2 &= \bar{V}_{n,RS}^2 \left(\frac{1}{2} \right)^2 \left(\frac{g_m R_F - 1}{1 + \frac{R_F}{R_0}} \right)^2 \end{aligned}$$

Noise of H_1 :

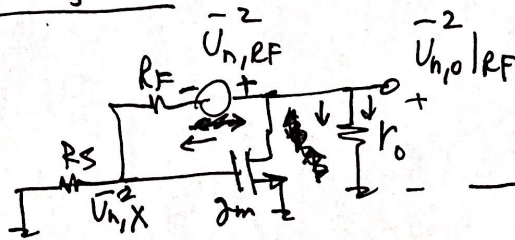


$$\frac{V_{n,RS}|_{M1}}{R_F + R_S} + \frac{V_{n,M1} \cdot g_m R_S}{R_F + R_S} + \frac{V_{n,M1}}{R_0} = \bar{V}_{n,M1}^2$$

$$V_{n,M1} \left[\frac{1}{R_0} + \frac{g_m R_S + 1}{R_F + R_S} \right] = \bar{V}_{n,M1}^2$$

$$\rightarrow V_{n,M1} = \bar{V}_{n,M1} \cdot \frac{1}{\frac{1}{R_0} + \frac{g_m R_S + 1}{R_F + R_S}}, \quad \bar{V}_{n,M1}^2 = \bar{V}_{n,M1}^2 \left[\frac{1}{\frac{1}{R_0} + \frac{g_m R_S + 1}{R_F + R_S}} \right]^2$$

Noise of R_F :



$$V_{n,RF} \left[\frac{1}{R_0} + \frac{1}{R_F} \right] = V_{n,RF} \left(\frac{1}{R_F} \right) + V_{n,X} \left(\frac{1}{R_F} \right) - g_m V_{n,X}$$

$$\text{Also, } V_{n,X} = (V_{n,RS}|_{RF} - V_{n,RF}) \cdot \frac{R_S}{R_F + R_S}$$

$$\rightarrow V_{n,RF} \left[\frac{1}{R_0} + \frac{1}{R_F + R_S} + \frac{g_m R_S}{R_F + R_S} \right] = V_{n,RF} \left[\frac{1}{R_F + R_S} + \frac{g_m R_S}{R_F + R_S} \right]$$

$$\overline{V_{nro}}^2|_{R_F} = \left[\frac{\frac{1}{R_F + R_S} [1 + g_m R_S]}{\frac{1}{r_o} + \frac{1}{R_F + R_S} (1 + g_m R_S)} \right]^2 \cdot \overline{V_{A|RF}}^2$$

$$NF = 1 + \frac{R_F}{R_S} \cdot \left[\frac{\frac{1}{R_F + R_S} (1 + g_m R_S)}{\frac{1}{r_o} + \frac{1}{R_F + R_S} (1 + g_m R_S)} \right]^2 \cdot \frac{4 \left(1 + \frac{R_F}{r_o}\right)^2}{(g_m R_F - 1)^2} \\ + \frac{\gamma g_m}{R_S} \left[\frac{1}{\frac{1}{r_o} + \frac{g_m R_S + 1}{R_F + R_S}} \right]^2 \cdot \frac{4 \left(1 + \frac{R_F}{r_o}\right)^2}{(g_m R_F - 1)^2} \quad (g_m R_F \gg 1)$$

→ input matching: $g_m \approx \frac{1 + R_F/r_o}{R_S}$ or $g_m R_S \approx 1 + \frac{R_F}{r_o}$

$$NF \approx 1 + \frac{R_F}{R_S} \left[\frac{\frac{1}{R_F + R_S} (1 + g_m R_S)}{\frac{1}{r_o} + \frac{1}{R_F + R_S} (1 + g_m R_S)} \right]^2 \cdot \frac{4 (g_m R_S)^2}{(g_m R_F)^2}$$

$$+ \frac{\gamma g_m}{R_S} \left[\frac{1}{\frac{1}{r_o} + \frac{1 + g_m R_S}{R_F + R_S}} \right]^2 \cdot \frac{4 (g_m R_S)^2}{(g_m R_F)^2}$$

$$NF \approx 1 + \frac{4 R_S}{R_F} \frac{(1 + g_m R_S)^2}{\left[\frac{R_F + R_S}{r_o} + (1 + g_m R_S) \right]^2} + \frac{4 \gamma g_m R_S}{R_F^2} \frac{[R_F + R_S]^2}{\left[\frac{R_F + R_S}{r_o} + (1 + g_m R_S) \right]^2}$$

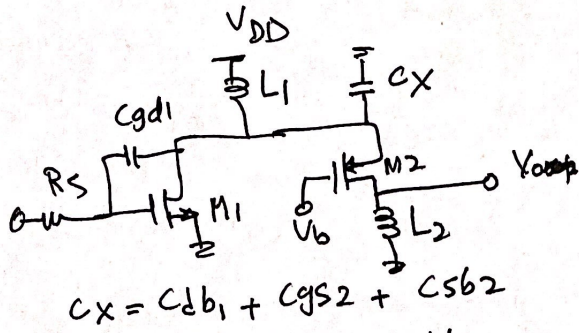
If $\left[\begin{array}{l} r_o \gg R_F, R_S \\ g_m R_S \approx 1 \end{array} \right] \rightarrow$

$$NF \approx 1 + \frac{4 R_S}{R_F} \frac{(1 + g_m R_S)^2}{(1 + g_m R_S)^2} + \frac{4 \gamma g_m R_S}{R_F^2} \frac{(R_F + R_S)^2}{(1 + g_m R_S)^2}$$

$$NF \approx 1 + \gamma + \frac{4 R_S}{R_F}$$

Problem 5.13 :

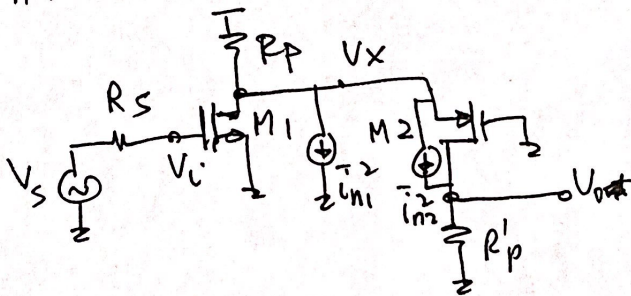
P.4



$$C_X = C_{db1} + C_{gs2} + C_{sb2}$$

→ Folded-Cascode LNA

• At resonance :



- for low supply voltage
- Neglect r_o , g_{mb}
- Neglect noise due to L_2 loss
- L_1 can be modeled at resonance as R_p
- L_2 loss is modeled as R_p'

$$V_X \approx -g_{m1} (R_p \parallel \frac{1}{g_{m2}}) V_i$$

$$V_i \approx V_S$$

$$V_o \approx g_{m2} R_p' \cdot V_X$$

$$\rightarrow V_o \approx -g_{m1} \frac{R_p}{1+g_{m2}R_p} \cdot g_{m2} R_p' V_i$$

$$\rightarrow V_o \approx -g_{m1} R_p' \left[\frac{g_{m2} R_p}{1+g_{m2}R_p} \right] V_i$$

$$\overline{i_{n1}^2} = 4KT\gamma g_{m1} + \frac{4KT}{R_p}, \quad \overline{i_{n2}^2} = 4KT\gamma g_{m2}$$

$$\overline{V_{on1}^2} = \overline{i_{n1}^2} \cdot \left[(R_p \parallel \frac{1}{g_{m2}}) \cdot (g_{m2} R_p') \right]^2 \approx \overline{i_{n1}^2} \cdot \left[\frac{g_{m2} R_p}{1+g_{m2}R_p} \cdot R_p' \right]^2$$

$$\overline{V_{on2}^2} = \frac{4KT\gamma}{g_{m2}} \cdot \left(\frac{R_p'}{R_p + \frac{1}{g_{m2}}} \right)^2 = \frac{4KT\gamma}{g_{m2}} \cdot \left(\frac{g_{m2} R_p'}{1+g_{m2}R_p} \right)^2 = \frac{4KT\gamma}{g_{m2}} \cdot \left(\frac{R_p' g_{m2}}{1+g_{m2}R_p} \right)^2$$

$$R_p \parallel \frac{1}{g_{m2}} = \frac{R_p / g_{m2}}{R_p + \frac{1}{g_{m2}}} = \frac{R_p}{1+g_{m2}R_p}$$

$$NF = 1 + \frac{(4KT\gamma g_{m1}) \cdot \left(\frac{g_{m2} R_p}{1+g_{m2}R_p} \right)^2 R_p'^2 + \left(\frac{4KT}{R_p} \right) \left(\frac{g_{m2} R_p}{1+g_{m2}R_p} \right)^2 R_p'^2}{4KT R_S g_{m1}^2 R_p'^2 \left(\frac{g_{m2} R_p}{1+g_{m2}R_p} \right)^2}$$

$$NF = 1 + g_{m2} 4KT\gamma \frac{R_p'^2}{(1+g_{m2}R_p)^2}$$

$$NF = 1 + \frac{\gamma}{g_{m1} R_S} + \frac{1}{(g_{m1} R_S)(g_{m1} R_P)} + \frac{\gamma}{g_{m1}^2 g_{m2} R_S R_P^2}$$

7.5

For $R_P \rightarrow \infty$, $NF \approx 1 + \frac{\gamma}{g_{m1} R_S}$ (ignoring R_P noise)

↓
Similar to a regular cascode LNA

Problem 5.16

NF of a CG stage with feedback while considering the noise of the feedback network $\overline{U_{NF}}^2$.

Input impedance:

$$Z_i = \frac{V_i}{i_i} = \frac{1}{g_m} + \alpha R_1$$

(@ resonance)

$$R_S = \frac{1}{g_m} + \alpha R_1$$

(matching condition)

Voltage gain: $\frac{V_o}{V_S} = \frac{1}{2} \frac{V_o}{V_{in}} = \frac{1}{2} \cdot \frac{R_1}{R_S}$

Noise of R_1 :

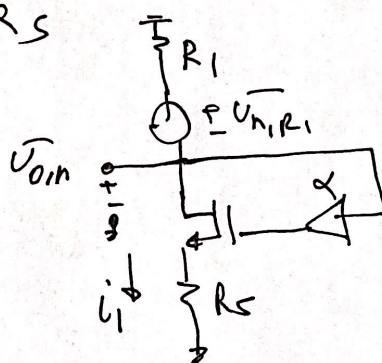
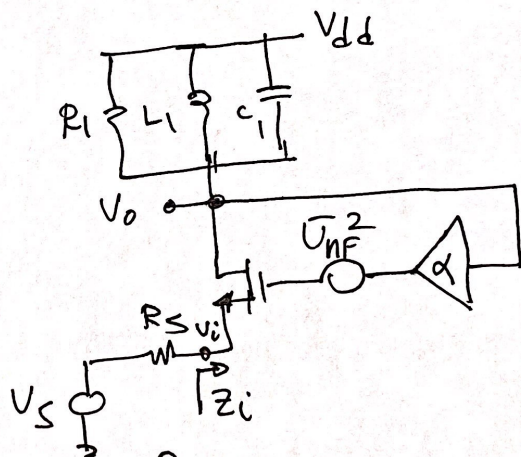
$$i_1 = (\alpha V_{on} - i_i R_S) g_m$$

$$i_1 = \frac{\alpha g_m V_{on}}{1 + g_m R_S}$$

$$\overline{V_{on}} = -i_1 R_1 + \overline{U_{n1} R_1}$$

$$\rightarrow \overline{V_{on}} = -R_1 \left(\frac{\alpha g_m V_{on}}{1 + g_m R_S} \right) + \overline{U_{n1} R_1}$$

$$\rightarrow \overline{V_{on}} = \frac{1 + g_m R_S}{1 + g_m R_S + \alpha g_m R_S} \overline{U_{n1} R_1} = \frac{1 + g_m R_S}{2 g_m R_S} \overline{U_{n1} R_1}$$



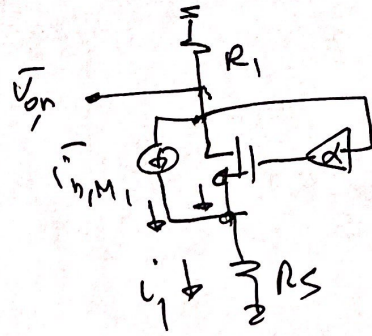
Noise of M_1 :

$$\bar{V}_{on} = -i_i R_1$$

$$i_i = i_{n,M_1} - g_m (i_i R_S + \alpha i_i R_1)$$

$$i_i = \frac{i_{n,M_1}}{1 + g_m R_S + \alpha g_m R_1}$$

$$\bar{V}_{on} = \frac{i_{n,M_1} R_1}{1 + g_m R_S + \alpha g_m R_1} = \frac{i_{n,M_1} R_1}{2 g_m R_S}$$



Noise of α :

$$V_{on} = -i_i R_1$$

$$i_i = g_m (V_{n,F} + \alpha V_{n,o} - i_i R_S)$$

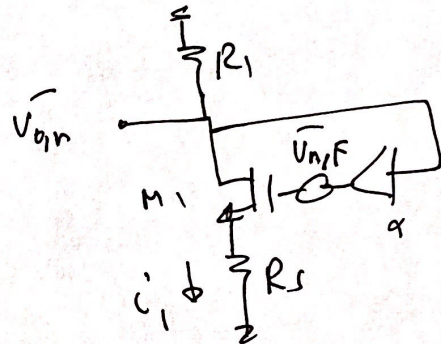
$$i_i = g_m (V_{n,F} - (\alpha R_1 + R_S) i_i)$$

$$\rightarrow i_i = \frac{g_m V_{n,F}}{1 + (\alpha R_1 + R_S) g_m} = \frac{g_m V_{n,F}}{g_m \left[\frac{1}{g_m} + \alpha R_1 + R_S \right]} = \frac{V_{n,F}}{2 R_S}$$

$$\bar{V}_{n,o} = \frac{-R_1}{2 R_S} \bar{V}_{n,F}$$

$$NF = 1 + \frac{4KTR_1 \left[\frac{1 + g_m R_S}{2 g_m R_S} \right]^2 + 4KTR_1 g_m \left(\frac{R_1}{2 g_m R_S} \right)^2 + \left(\frac{R_1 \bar{V}_{n,F}^2}{4 R_S^2} \right)}{4KTR_S \left[\frac{1}{2} \cdot \frac{R_1}{R_S} \right]^2}$$

$$NF = 1 + \frac{R_S}{g_m R_S} + \frac{R_S}{R_1} \left(1 + \frac{1}{g_m R_S} \right)^2 + \frac{\bar{V}_{n,F}^2}{4KTR_S}$$



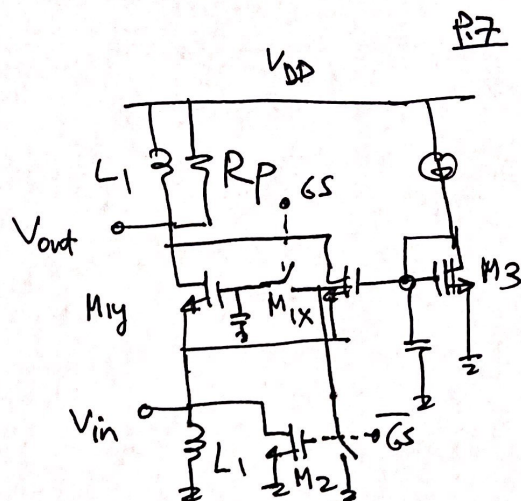
Problem 5.21

Gain switching

For maximum gain

M_{1x}, M_{1y} are on, M_2 is off

$$\text{Gain} \approx (g_{m1x} + g_{m1y}) \cdot R_p$$



For 3dB lower

$$\text{Gain} \approx Z_{m1} \times R_p$$

(M_1 is off, M_2 is on)

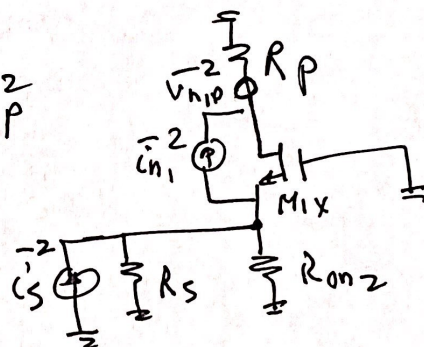
$$\frac{w_{ix}}{w_{ix} + w_{iy}} = \frac{1}{\sqrt{2}} \quad , \quad R_{on2} = \frac{\sqrt{2}}{\sqrt{2}-1} R_S$$

$$R_5 \parallel R_{on2} = \frac{1}{\sqrt{2} R_5}$$

Noise Analysis

$$V_{o, n, \text{full}}^{-2} \left(\frac{4kT}{R_S} + \frac{4kT}{R_{on2}} \right) (R_S \parallel R_{on2})^2 \times g_{m1}^2 R_P^2$$

$$+ 4kTR_p + \frac{4kT\gamma}{g_{m1}} \cdot \frac{R_p^2}{\left[\frac{1}{g_{m1}} + (R_S \parallel R_{on2})^2 \right]}$$

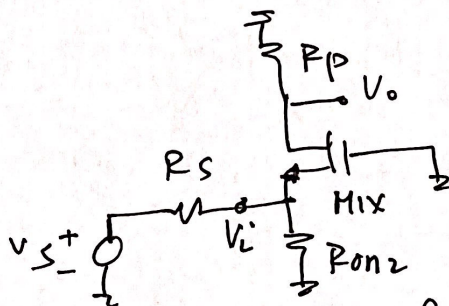


$$\left(\frac{-1.2}{15} = \frac{4KT}{R_S} + \frac{4KT}{R_{on2}} \right)$$

$$NF = \frac{V_{on, total}^2}{4kTR_s \cdot A_v^2}$$

$$A_v = \frac{V_o}{V_s} = \frac{R_{on2}}{R_{on2} + R_s} \cdot g_{m1} R_D$$

$$A_v = \frac{g_{m1} R_p}{2}$$



$$\rightarrow NF = 1 + \frac{R_S}{R_{on2}} + \frac{R_p R_S}{(R_S \parallel R_{on2})^2 g_{m_x, p}^2} + \frac{g_{m_x, s} R_S}{(1 + g_{m_x, s} (R_S \parallel R_{on2}))^2 \cdot (R_{S \parallel on2})^2 \cdot g_{m_x, s}^2}$$

$$NF = 1 + \left(\frac{\sqrt{2} - 1}{\sqrt{2}} \right) + \frac{4}{(1/\sqrt{2})^2} \cdot \frac{R_S}{R_p} + \frac{8/\sqrt{2}}{\left[1 + \frac{1}{2\sqrt{2} - 1} \right]^2 \left(\frac{1}{2\sqrt{2}} \right)^2}$$

$$NF \approx 1 + 0.293 + 2.368 + 8 \frac{R_S}{R_p} = 1.293 + 2.368 + 8 \frac{R_S}{R_p}$$